Table II: Comparative Evaluation of Baseline Models vs. Proposed Framework

Model: Multi-Commodity Random Forest Regressor | Evaluation: 5-Fold Expanding Walk-Forward Time-Series Cross-Validation (Zero Lookahead Bias)

Model	Validation Strategy	Fold 1 R^2 / MAPE	Fold 2 R^2 / MAPE	Fold 3 R^2 / MAPE	Fold 4 R^2 / MAPE	Fold 5 R^2 / MAPE	Mean R^2	Mean MAPE	Std R^2 (Stability)	Mean M (Rs./Qt
ARIMA	Walk-Forward	0.6665 / 33.37%	0.7926 / 21.65%	0.8951 / 22.80%	0.6638 / 21.58%	0.8485 / 22.51%	0.7733	24.38%	0.0941	Rs. 469
XGBoost	Walk-Forward	0.8663 / 10.55%	0.9442 / 8.94%	0.9652 / 7.00%	0.9217 / 8.42%	0.8589 / 8.96%	0.9113	8.77%	0.0421	Rs. 198
(Proposed)	Walk-Forward	**0.9354 / 8.33%**	**0.9527 / 7.90%**	**0.9660 / 6.92%**	**0.9408 / 8.16%**	**0.9495 / 7.90%**	**0.9489**	**7.84%**	**0.0105**	**Rs. 155**

Findings: (1) Proposed RF outperforms ARIMA by **16.54% MAPE** (24.38% → 7.84%). (2) RF outperforms XGBoost by **0.93% MAPE** (8.77% → 7.84%). (3) RF is **4.0x more stable** than XGBoost (Std 05 vs 0.0421). (4) Wilcoxon p-value RF vs ARIMA = 0.0625 — consistent with small-sample limitation (n=5 folds); magnitude of improvement is unambiguous.

Table III: Per-Fold Detailed Results — Random Forest (Proposed Method)

Fold #	Training Period	Validation Period	Train Records	Val Records	R^2 Score	MAE (Rs./Qtl)	RMSE	MAPE (%)	Accuracy
Fold 1	2001–2010	2011–2013	349,869	177,370	**0.9354**	Rs. 145.84	304.86	**8.33%**	**91.67%**
Fold 2	2001–2013	2014–2016	527,239	179,715	**0.9527**	Rs. 126.85	269.78	**7.90%**	**92.10%**
Fold 3	2001–2016	2017–2018	706,954	181,787	**0.9660**	Rs. 111.22	239.50	**6.92%**	**93.08%**
Fold 4	2001–2018	2019–2021	888,741	235,460	**0.9408**	Rs. 184.17	395.13	**8.16%**	**91.84%**
Fold 5	2001–2021	2022–2024	1,124,201	189,906	**0.9495**	Rs. 211.68	470.56	**7.90%**	**92.10%**
Mean	**2001–Expanding**	**2011–2024**	**729,401 avg**	**192,848 avg**	**0.9489**	**Rs. 155.95**	**335.97**	**7.84%**	**92.16%**